

6(a)	energy (transferred from electrical to other forms) per unit charge	B1
6(b)(i)	(resistance is) infinite / very high	B1
6(b)(ii)	(resistance) decreases (as V increases)	B1
6(c)(i)	current = $2.7 - 1.5$ $= 1.2 \text{ A}$	A1
6(c)(ii)	$12 = (1.5 \times 5.0) + (1.5 \times R)$ or $R = (12/1.5) - 5.0$	C1
	$R = 3.0 \Omega$	A1
6(c)(iii)	$V_{(XZ)} = (1.6 / 2.0) \times 12 (= 9.6 \text{ V})$	C1
	$V_{(XW)} = 1.5 \times 5.0 (= 7.5 \text{ V})$	C1
	potential difference = $9.6 - 7.5$ $= 2.1 \text{ V}$	A1
	or	
	$V_{(ZY)} = (0.4 / 2.0) \times 12 (= 2.4 \text{ V})$	(C1)
	$V_{(WY)} = 1.5 \times 3.0 (= 4.5 \text{ V})$	(C1)
	potential difference = $4.5 - 2.4$ $= 2.1 \text{ V}$	(A1)

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Question	Answer	Marks
6(c)(iv)	current in (fixed / variable) resistor decreases	B1
	current in (resistance) wire is unchanged	B1
	(so) current in battery decreases, (same e.m.f. so) power decreases	B1

Question	Answer	Marks
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